



COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING

FOR THE COURSE
OPERATING SYSTEMS

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CE-46-A

Multithreaded University Course Enrollment System

Introduction

Background

In modern universities, course registration systems must handle multiple students trying to enroll at the same time. This creates a resource-sharing problem, where limited seats must be distributed efficiently and fairly.

This scenario closely resembles classical Operating System problems involving:

- Process scheduling
- Resource allocation
- Synchronization
- Deadlock avoidance

Motivation

The motivation behind this project was to apply core OS concepts in a practical system instead of only studying them theoretically.

Course enrollment is an ideal real-world example because:

- Students behave like processes
- Courses behave like resources
- Seats represent limited resource units
- Simultaneous requests create race conditions

System Overview

This project simulates a multithreaded enrollment system where:

- Each student request is handled as a thread
- Threads compete for limited seats
- Execution is controlled using priority scheduling
- Shared data is protected using mutex locks
- System safety is ensured using Banker's Algorithm

Problem Statement

Design a system that allows multiple students to enroll in courses concurrently while ensuring:

- No over-allocation of seats
- No race conditions
- No unsafe states (deadlocks)
- Fair scheduling based on priority

Challenges + Solutions

- Multiple threads accessing shared data → race conditions
- Limited seats → resource contention
- Enrollment constraints → complex validation
- Concurrent execution → non-deterministic behavior

Objectives

Primary Objective

To develop a safe and concurrent enrollment system using Operating System techniques.

Secondary Objectives

Implement multithreading using C++
Apply mutex locks for synchronization
Use condition variables for scheduling
Implement Banker's Algorithm for safety
Enforce real-world enrollment constraints

Learning Outcomes

Understanding of thread synchronization
Practical use of deadlock avoidance
Experience with shared resource handling
Insight into scheduling techniques

System Overview

Workflow

1. System initializes courses and students
2. User logs in
3. Student selects department and course
4. System performs validation checks
5. Banker's Algorithm ensures safe allocation
6. Threads are created for competing students
7. Scheduler controls execution order
8. Enrollment occurs in critical section
9. Results are displayed and saved

Design Rationale

The system follows a stepwise pipeline to:

Ensure correctness before concurrency

Avoid unnecessary thread creation

Maintain system stability

System Design and Justification

Multithreading

Each student request is represented as a thread.

Simulates real-world concurrent access where multiple users interact with the system simultaneously.

Shared Data Structures

Global vectors store:

Students

Courses

Reason:

All threads must access the same data to reflect a real shared system.

Mutex for Critical Section

A mutex lock is used during enrollment updates.

Prevents:

Overfilling seats

Data corruption

Inconsistent updates

Priority Scheduling

Threads are sorted based on student priority.

Simulates real-world policies (e.g., seniors get priority).

Condition Variable (Turn-Based Execution)

Threads execute only when their turn arrives.

Ensures:

Controlled execution

Fair scheduling

Avoids random thread execution order

Algorithmic Design

Resource Allocation Model

Students → Processes

Courses → Resources

Seats → Resource units

Banker's Algorithm

Purpose: Prevent unsafe states

Before granting a request:

System checks if all students can still complete enrollment

Reason for using it:

Ensures future safety, not just current availability

Prevents deadlock-like conditions

Scheduling Strategy

Requests sorted by priority

Executed using:

 `currentTurn`

 `condition\_variable`

Reason:

Simulates OS-level priority scheduling

Implementation Details

Language

C++

Libraries Used

`<thread>` → multithreading

`<mutex>` → synchronization

`<condition\_variable>` → scheduling

`<atomic>` → shared control

`<fstream>` → file handling

Key Functions

enrollThread()

- * Represents a student thread
- * Waits for its turn
- * Executes critical section
- * Updates enrollment

bankersAlgorithmSafe()

- * Checks if system remains safe
- * Prevents unsafe allocation

enroll()

- * Controls entire enrollment process
- * Performs validation and thread creation

dropCourse()

- * Removes enrollment
- * Frees seat safely

Output Explanation

When the program runs, it displays thread execution logs that simulate concurrent enrollment.

Sample Output

[THREAD 3] Alice (Priority 4) - ENROLLED (Seats filled: 5/30)
[THREAD 7] Bob (Priority 2) - BLOCKED (semaphore blocked - no seats)
[THREAD 1] Diana (Priority 5) - ENROLLED (Seats filled: 6/30)

Thread Identifier

[THREAD 3]

Represents a separate execution unit (process simulation)

Each thread corresponds to one student request

Student Information

Alice (Priority 4)

Shows student name and priority level

Higher priority; executed earlier

Status Messages

ENROLLED (Seats filled: 5/30)

Enrollment successful

Seat count updated

Student record modified

Saved to file

BLOCKED (semaphore blocked - no seats)

Possible reasons include:

No available seats

Maximum course limit reached

Already enrolled

Unsafe state detected by Banker's Algorithm

Critical Section Behavior

Only one thread executes enrollment at a time.

Protected using mutex

Prevents race conditions

Scheduling Behavior

Threads do NOT execute randomly
They follow priority order

Example:

Priority 5 executes before Priority 3

Banker's Algorithm in Output

If unsafe:

BLOCKED (Banker's: unsafe state)

Granting this request would lead to unsafe system state

Enrollment is prevented

Final Result Section

After all threads finish:

FINAL ENROLLMENT RESULT FOR: Alice

>> ENROLLMENT SUCCESSFUL <<

Course: CS201 - Data Structures

OR

>> ENROLLMENT REJECTED <<

Reason: max courses reached

Key Observations

No seat over-allocation

No duplicate enrollments

Fair execution order

System remains safe

Sample Outputs of Code

```
Microsoft Visual Studio Debug Console
=====
UNIVERSITY COURSE ENROLLMENT SYSTEM
Multithreaded | Priority Scheduling | Banker's
Algorithm | Mutex | Semaphore
Max 6 Courses: 5 Dept + 1 Elective
=====
Enter Full Name : Maryam
Enter Student ID: 1007

Welcome, Maryam! (Department: Mechanical | Priority: 4)
=====
WHAT WOULD YOU LIKE TO DO?
=====
1. Enroll in a course
2. Drop a course
3. View enrollment summary
4. Exit
=====
Enter choice (1-4): 1
=====
ENROLLMENT SUMMARY FOR: Maryam
=====
Total courses enrolled : 0 / 6
Dept courses           : 0 / 5
Elective courses        : 0 / 1
=====

SELECT A DEPARTMENT
=====
1. Computer Science
2. Electrical Engineering
3. Mechanical Engineering
4. Mechatronics
=====
Enter choice (1-4): 3
=====
COURSES : Mechanical
=====
#   Code   Title                               Available Seats
=====
0   ME101   Engineering Mechanics                       30
1   ME201   Thermodynamics                             25
2   ME301   Fluid Mechanics                             20
3   ME401   Machine Design                             20
4   ME501   Manufacturing Processes                     15
=====
Select course (0-4): 3
You selected: [ME401] Machine Design
=====
PRE-ENROLLMENT CHECKS
=====
No conflicts found. Running Banker's Algorithm...
=====
BANKER'S ALGORITHM : SAFETY CHECK
=====
Max demand per student : 6 courses
(5 from own dept + 1 elective from any other dept)

Safe sequence found. System is in a SAFE STATE.
Proceeding to thread scheduling...
=====

THREAD SIMULATION : CONCURRENT ENROLLMENT
=====
Spawning 5 threads for 20 available seat(s)...
Mutex locks critical section. Semaphore tracks seats.
Priority scheduler releases threads highest-first.
Rules: max 6 total | max 5 dept | max 1 elective
```

```
[THREAD 2 ] Javeria          (Priority 4) - spawned, waiting for scheduler...
[THREAD 1 ] Danish           (Priority 4) - spawned, waiting for scheduler...
[THREAD 4 ] Zeeshan          (Priority 4) - spawned, waiting for scheduler...
[THREAD 5 ] Maryam           (Priority 4) - spawned, waiting for scheduler...
[THREAD 3 ] Maheen           (Priority 4) - spawned, waiting for scheduler...
[THREAD 5 ] Maryam           (Priority 4) - ENROLLED (Seats filled: 1/20)
[THREAD 2 ] Javeria          (Priority 4) - ENROLLED (Seats filled: 2/20)
[THREAD 3 ] Maheen           (Priority 4) - ENROLLED (Seats filled: 3/20)
[THREAD 4 ] Zeeshan          (Priority 4) - ENROLLED (Seats filled: 4/20)
[THREAD 1 ] Danish           (Priority 4) - ENROLLED (Seats filled: 5/20)
```

```
=====
FINAL ENROLLMENT RESULT FOR: Maryam
=====
```

```
-----
>> ENROLLMENT SUCCESSFUL <<
Course : ME401 - Machine Design
Dept   : Mechanical
Type   : Department Course
Record saved to enrollments.txt
-----
```

```
Updated enrollment status:
Total: 1/6 | Dept: 1/5 | Elective: 0/1
-----
```

```
=====
Continue? (y/n): y
=====
```

```
=====
WHAT WOULD YOU LIKE TO DO?
=====
```

1. Enroll in a course
2. Drop a course
3. View enrollment summary
4. Exit

```
=====
Enter choice (1-4): 1
=====
```

```
=====
ENROLLMENT SUMMARY FOR: Maryam
=====
```

```
Total courses enrolled : 1 / 6
Dept courses           : 1 / 5
Elective courses       : 0 / 1
Enrolled in            : ME401 (Machine Design)
=====
```

```
=====
SELECT A DEPARTMENT
=====
```

1. Computer Science
2. Electrical Engineering
3. Mechanical Engineering
4. Mechatronics

```
=====
Enter choice (1-4): 3
=====
```

```

=====
COURSES : Mechanical
=====
#   Code   Title                                     Available Seats
-----
0   ME101   Engineering Mechanics                     30
1   ME201   Thermodynamics                               25
2   ME301   Fluid Mechanics                                   20
3   ME401   Machine Design                                    15
4   ME501   Manufacturing Processes                            15
=====
Select course (0-4): 2

You selected: [ME301] Fluid Mechanics

=====
PRE-ENROLLMENT CHECKS
=====
No conflicts found. Running Banker's Algorithm...

=====
BANKER'S ALGORITHM : SAFETY CHECK
=====
Max demand per student : 6 courses
(5 from own dept + 1 elective from any other dept)

Safe sequence found. System is in a SAFE STATE.
Proceeding to thread scheduling...

=====

=====
THREAD SIMULATION : CONCURRENT ENROLLMENT
=====
Spawning 5 threads for 20 available seat(s)...
Mutex locks critical section. Semaphore tracks seats.
Priority scheduler releases threads highest-first.
Rules: max 6 total | max 5 dept | max 1 elective

[THREAD 3 ] Zeeshan      (Priority 4) - spawned, waiting for scheduler...
[THREAD 1 ] Maheen      (Priority 4) - spawned, waiting for scheduler...
[THREAD 2 ] Javeria     (Priority 4) - spawned, waiting for scheduler...
[THREAD 5 ] Danish      (Priority 4) - spawned, waiting for scheduler...
[THREAD 4 ] Maryam      (Priority 4) - spawned, waiting for scheduler...
[THREAD 4 ] Maryam      (Priority 4) - ENROLLED (Seats filled: 1/20)
[THREAD 2 ] Javeria     (Priority 4) - ENROLLED (Seats filled: 2/20)
[THREAD 1 ] Maheen      (Priority 4) - ENROLLED (Seats filled: 3/20)
[THREAD 3 ] Zeeshan     (Priority 4) - ENROLLED (Seats filled: 4/20)
[THREAD 5 ] Danish      (Priority 4) - ENROLLED (Seats filled: 5/20)

=====
FINAL ENROLLMENT RESULT FOR: Maryam
=====

>> ENROLLMENT SUCCESSFUL <<
Course : ME301 - Fluid Mechanics
Dept   : Mechanical
Type   : Department Course
Record saved to enrollments.txt

-----
Updated enrollment status:
Total: 2/6 | Dept: 2/5 | Elective: 0/1
-----

Continue? (y/n): y

```

```

=====
WHAT WOULD YOU LIKE TO DO?
=====
1. Enroll in a course
2. Drop a course
3. View enrollment summary
4. Exit
=====
Enter choice (1-4): 1

=====
ENROLLMENT SUMMARY FOR: Maryam
=====
Total courses enrolled : 2 / 6
Dept courses           : 2 / 5
Elective courses       : 0 / 1
Enrolled in            : ME401 (Machine Design), ME301 (Fluid Mechanics)
=====

=====
SELECT A DEPARTMENT
=====
1. Computer Science
2. Electrical Engineering
3. Mechanical Engineering
4. Mechatronics
=====
Enter choice (1-4): 2

[NOTE] You are selecting a course outside your department.
This will count as your elective (max 1 allowed).

=====
COURSES : Electrical
=====
#   Code   Title                               Available Seats
-----
0   EE101   Circuit Analysis                       30
1   EE201   Electronics                                   25
2   EE301   Signals & Systems                            20
3   EE401   Power Systems                                20
4   EE501   Control Systems                             15
=====
Select course (0-4): 1

You selected: [EE201] Electronics

=====
PRE-ENROLLMENT CHECKS
=====
No conflicts found. Running Banker's Algorithm...

=====
BANKER'S ALGORITHM : SAFETY CHECK
=====
Max demand per student : 6 courses
(5 from own dept + 1 elective from any other dept)

Safe sequence found. System is in a SAFE STATE.
Proceeding to thread scheduling...
=====

=====
THREAD SIMULATION : CONCURRENT ENROLLMENT
=====
Spawning 6 threads for 25 available seat(s)...
Mutex locks critical section. Semaphore tracks seats.
Priority scheduler releases threads highest-first.
Rules: max 6 total | max 5 dept | max 1 elective

```

```
[THREAD 2 ] Adnan          (Priority 3) - spawned, waiting for scheduler...
[THREAD 1 ] Maryam        (Priority 4) - spawned, waiting for scheduler...
[THREAD 3 ] Bilal         (Priority 3) - spawned, waiting for scheduler...
[THREAD 1 ] Maryam        (Priority 4) - ENROLLED (Seats filled: 1/25)
[THREAD 2 ] Adnan         (Priority 3) - ENROLLED (Seats filled: 2/25)
[THREAD 5 ] Fawad         (Priority 3) - spawned, waiting for scheduler...
[THREAD 4 ] Arooj         (Priority 3) - spawned, waiting for scheduler...
[THREAD 6 ] Rehan         (Priority 3) - spawned, waiting for scheduler...
[THREAD 3 ] Bilal         (Priority 3) - ENROLLED (Seats filled: 3/25)
[THREAD 5 ] Fawad         (Priority 3) - ENROLLED (Seats filled: 4/25)
[THREAD 4 ] Arooj         (Priority 3) - ENROLLED (Seats filled: 5/25)
[THREAD 6 ] Rehan         (Priority 3) - ENROLLED (Seats filled: 6/25)
```

```
=====
FINAL ENROLLMENT RESULT FOR: Maryam
=====
```

```
-----
>> ENROLLMENT SUCCESSFUL <<
Course : EE201 - Electronics
Dept   : Electrical
Type    : Elective
Record saved to enrollments.txt
-----
```

```
Updated enrollment status:
Total: 3/6 | Dept: 2/5 | Elective: 1/1
-----
```

```
=====
Continue? (y/n): y
```

```
=====
WHAT WOULD YOU LIKE TO DO?
=====
```

1. Enroll in a course
2. Drop a course
3. View enrollment summary
4. Exit

```
=====
Enter choice (1-4): 2
```

```
=====
ENROLLMENT SUMMARY FOR: Maryam
=====
```

```
Total courses enrolled : 3 / 6
Dept courses           : 2 / 5
Elective courses       : 1 / 1
Enrolled in            : ME401 (Machine Design), ME301 (Fluid Mechanics), EE201 (Electronics)
=====
```

```
=====
SELECT COURSE TO DROP
=====
```

0. [ME401] Machine Design (Mechanical)
1. [ME301] Fluid Mechanics (Mechanical)
2. [EE201] Electronics (Electrical)

```
=====
Enter index to drop (-1 to cancel): 1
```

```
You are about to drop: [ME301] Fluid Mechanics
Confirm? (y/n): y
```

```
=====
COURSE DROPPED SUCCESSFULLY
=====
```

```
Dropped : [ME301] Fluid Mechanics
Seat has been freed back to the course pool.
```

```
Updated enrollment status:
Total: 2/6 | Dept: 1/5 | Elective: 1/1
```

Continue? (y/n): y

=====

WHAT WOULD YOU LIKE TO DO?

=====

1. Enroll in a course
 2. Drop a course
 3. View enrollment summary
 4. Exit
- =====

Enter choice (1-4): 3

=====

ENROLLMENT SUMMARY FOR: Maryam

=====

Total courses enrolled : 2 / 6
Dept courses : 1 / 5
Elective courses : 1 / 1
Enrolled in : ME401 (Machine Design), EE201 (Electronics)

=====

Continue? (y/n): y

Continue? (y/n): y

=====

WHAT WOULD YOU LIKE TO DO?

=====

1. Enroll in a course
 2. Drop a course
 3. View enrollment summary
 4. Exit
- =====

Enter choice (1-4): 4

=====

Session ended. Final enrollment summary:

=====

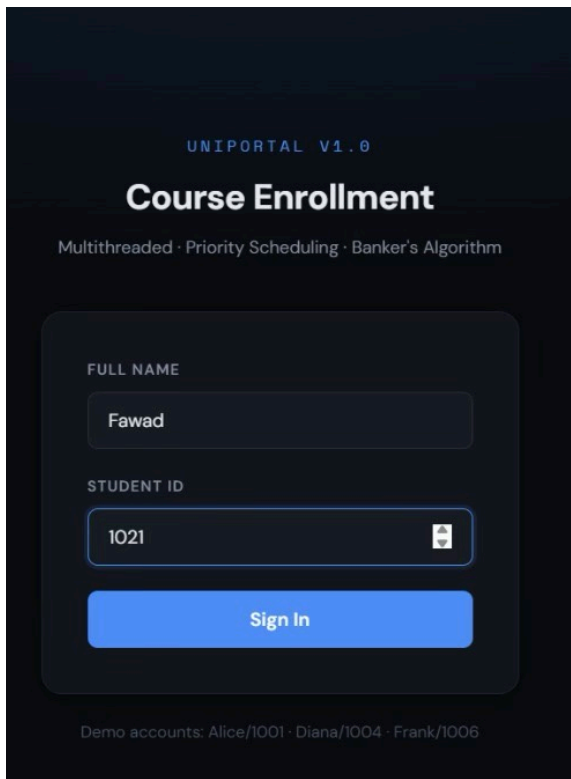
Student : Maryam
Total : 2 / 6
Dept : 1 / 5
Elective : 1 / 1
Courses : ME401 (Machine Design)
 EE201 (Electronics)

=====

C:\Users\P R O B O O K\Desktop\OS\Project\CEP\x64\Debug\CEP.exe (process 12132) exited
To automatically close the console when debugging stops, enable Tools->Options->Debuggi
Press any key to close this window . . .|

Sample Outputs of Code After Implementing GUI

Input :



UNIPORTAL V1.0

Course Enrollment

Multithreaded · Priority Scheduling · Banker's Algorithm

FULL NAME

Fawad

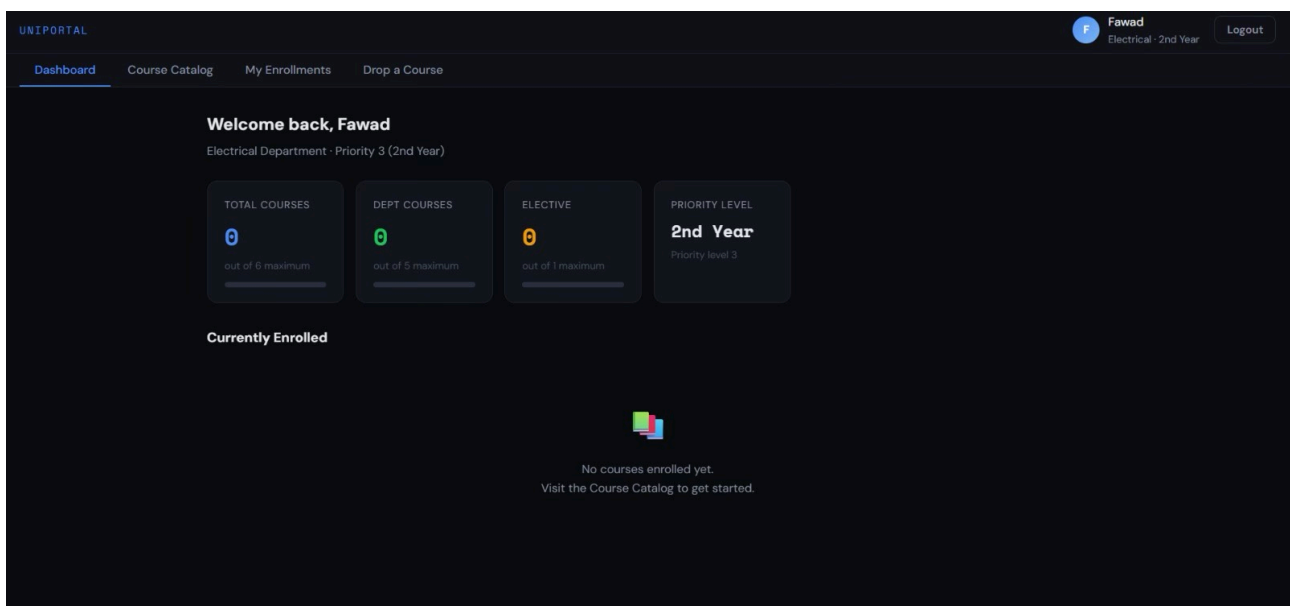
STUDENT ID

1021

Sign In

Demo accounts: Alice/1001 · Diana/1004 · Frank/1006

Student Dashboard before enrollment :



UNIPORTAL

Fawad
Electrical · 2nd Year

Logout

Dashboard Course Catalog My Enrollments Drop a Course

Welcome back, Fawad

Electrical Department · Priority 3 (2nd Year)

TOTAL COURSES
0
out of 6 maximum

DEPT COURSES
0
out of 5 maximum

ELECTIVE
0
out of 1 maximum

PRIORITY LEVEL
2nd Year
Priority level 3

Currently Enrolled

No courses enrolled yet.
Visit the Course Catalog to get started.

Course Catalog :

UNIportal

Fawad
Electrical - 2nd Year

Logout

Dashboard

Course Catalog

My Enrollments

Drop a Course

Course Catalog

Select a department to browse courses. Click Enroll to request a seat.

AllComputer ScienceElectricalMechanicalMechatronics

CS101

26 left

Intro to Programming

Computer · Elective

Seats filled4/30

Enroll as Elective

CS201

21 left

Data Structures

Computer · Elective

Seats filled4/25

Enroll as Elective

CS301

15 left

Operating Systems

Computer · Elective

Seats filled5/20

Enroll as Elective

CS401

15 left

Computer Networks

Computer · Elective

Seats filled5/20

Enroll as Elective

CS501

10 left

Artificial Intelligence

Computer · Elective

Seats filled5/15

Enroll as Elective

EE101

30 left

Circuit Analysis

Electrical

Seats filled0/30

Enroll

Status Message Display :

Thread Simulation

Close

Enrolling in EE201 — Electronics · Priority scheduling · Mutex · Semaphore

[THREAD 5]

Adnan

P3

BLOCKED

already enrolled in this course

[THREAD 4]

Bilal

P3

ENROLLED

(3/25)

[THREAD 2]

Fawad YOU

P3

ENROLLED

(4/25)

[THREAD 3]

Arooj

P3

ENROLLED

(5/25)

[THREAD 1]

Rehan

P3

ENROLLED

(6/25)

☒

Enrollment Successful

Your seat has been reserved and saved.

My Enrollments :

UNIportal

Fawad
Electrical - 2nd Year

Logout

Dashboard

Course Catalog

My Enrollments

Drop a Course

My Enrollments

All courses you are currently enrolled in this semester.

EE201

Electronics

Electrical

Dept

EE101

Circuit Analysis

Electrical

Dept

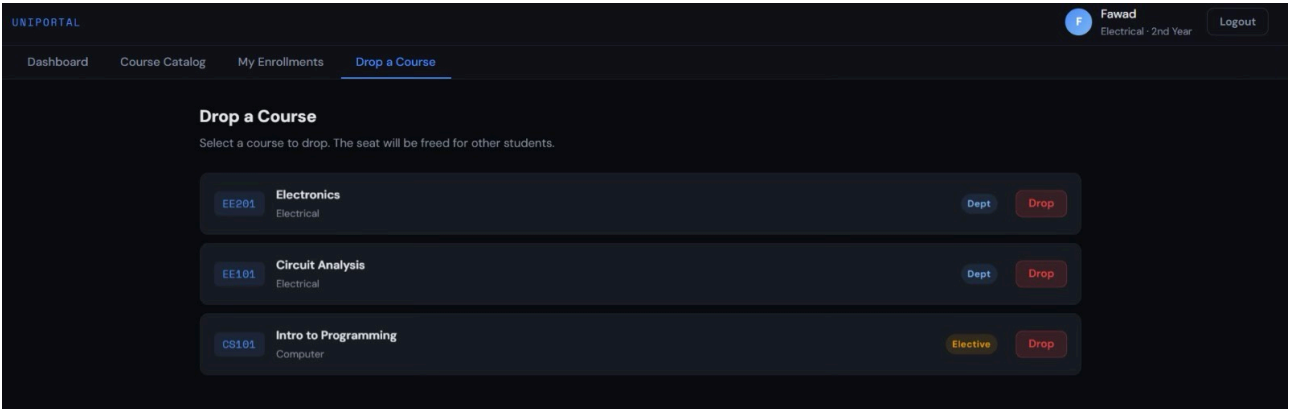
CS101

Intro to Programming

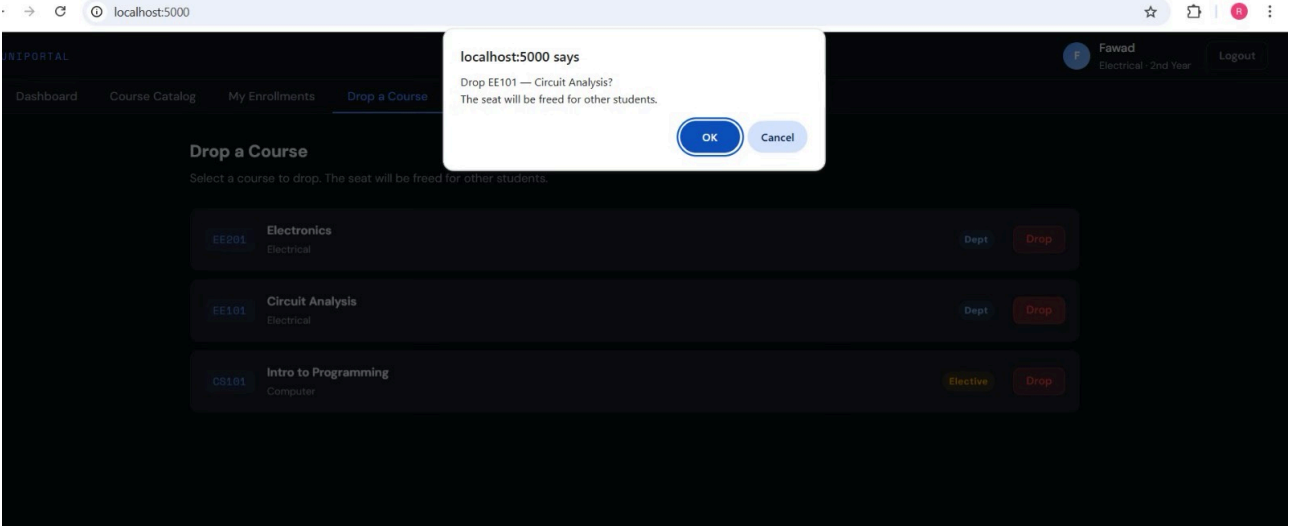
Computer

Elective

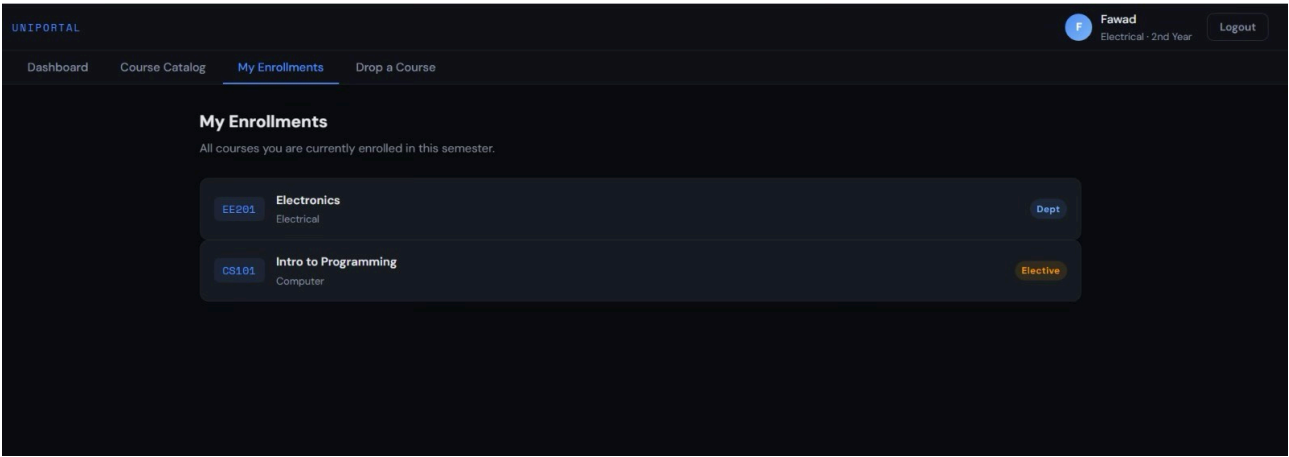
Drop a Course :



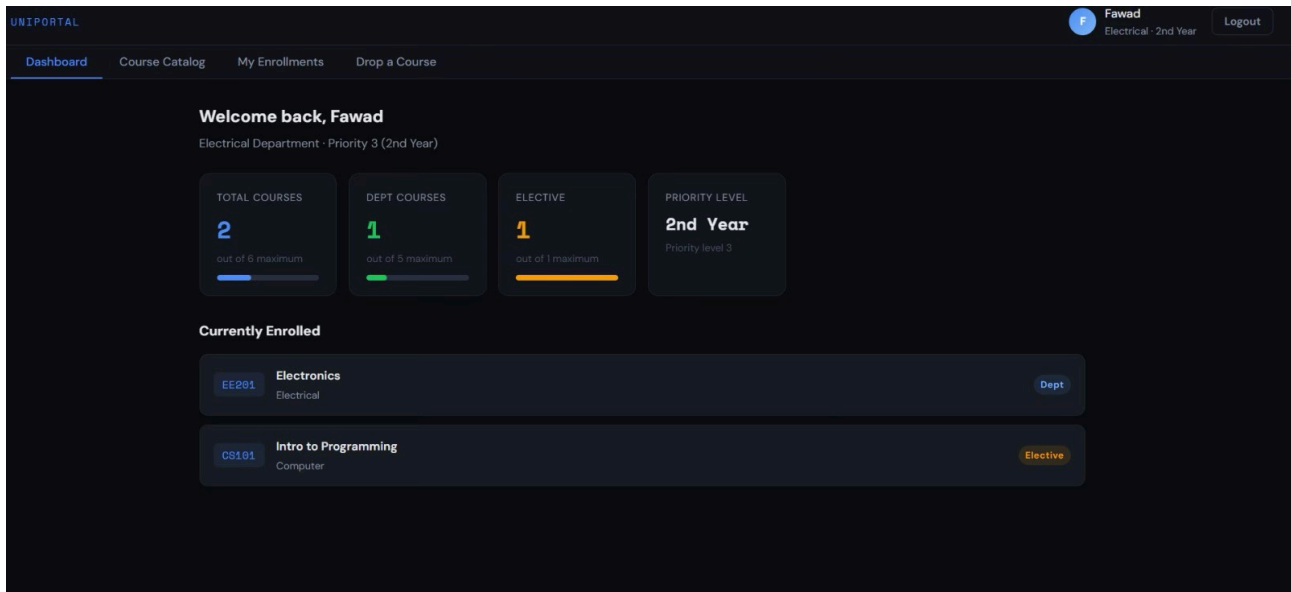
Process to Drop :



After dropping :



Student Dashboard after enrollment :



Results and Analysis

Observed Behavior :

- Threads executed in priority order
- Seat allocation remained accurate
- No race conditions occurred
- Deadlocks avoided successfully

System Reliability

Stable under concurrent requests
Correct enforcement of constraints

Challenges and Solutions

- Race Conditions

Solved using → Mutex

- Thread Coordination

Solved using → Condition Variable

- Deadlock Prevention

Solved using → Banker's Algorithm

- Data Consistency

Solved using → Global validation + controlled updates

Conclusion

This project successfully demonstrates the practical application of Operating System concepts in a real-world simulation.

The system:

Handles concurrent requests

Maintains data integrity

Prevents unsafe states

Ensures fairness

Future Enhancements

Database integration

Web-based system

Advanced scheduling algorithms

Multi-course parallel enrollment

References

Operating System Concepts (Silberschatz)

C++ Documentation

Online threading resources